Mixture-of-Flow velocity decomposition:

capacity ceiling vs. optimization gap, the posterior-responsibility form of a multi-domain teacher velocity, and when a MoF can be split into valid sub-flows

Flow Factory

Abstract

We formalize the Mixture-of-Flow (MoF) student for cross-model on-policy distillation (XOPD): a set of full flow-matching experts whose output velocities are blended by a router. We prove that the marginal velocity of a *mixture-of-domains* teacher is *exactly* a posterior-responsibility weighted convex combination of the per-domain velocities (Thm. 1), which pins down the "correct" functional form of a MoF student and explains why a sum-to-one (convex) gate makes every expert an individually valid flow. We then resolve an apparent paradox ("big→MoF→single beats big→single") by separating the *representational ceiling* (capacity-bound, path-independent) from the *optimization gap* (path-dependent). Finally we state when a mixture with fixed *activated* capacity can exceed a single model of the same activated capacity, and contrast MoF with classic (FFN-level) MoE. Companion progress note: `docs/xopd/progress/2026-07-21-mof-velocity-decomposition.md`.

1 Setup and notation

Flow matching learns a time-dependent velocity field $\boldsymbol{u}_t : \mathbb{R}^d \to \mathbb{R}^d$, $t \in [0,1]$, whose ODE $\dot{\boldsymbol{x}}_t = \boldsymbol{u}_t(\boldsymbol{x}_t)$ transports a base density p_0 (noise) to the data density p_1. For a probability path (p_t) induced by a coupling with per-sample conditional velocity $\boldsymbol{u}_t(\boldsymbol{x} \mid z)$ (where z is the data-side latent/condition), the *marginal* velocity is a conditional expectation

$$\boldsymbol{u}_t(\boldsymbol{x}) = \mathbb{E}[\dot{\boldsymbol{x}}_t \mid \boldsymbol{x}_t = \boldsymbol{x}] = \int \boldsymbol{u}_t(\boldsymbol{x} \mid z)\, p_t(z \mid \boldsymbol{x})\, \mathrm{d}z. \tag{1}$$

The map "target ↦ marginal velocity" is **linear** (it is an expectation); this is the single structural fact behind everything below.

Definition 1 (Valid flow)**.** *A field $\boldsymbol{u}_t$ is a* valid flow *for a path (p_t) if it satisfies the continuity equation $\partial_t p_t + \nabla\cdot(p_t\, \boldsymbol{u}_t) = 0$ with boundary $p_0 =$ noise, $p_1 =$ data; equivalently it is the marginal velocity* (1) *of some coupling reaching p_1.*

2 The teacher velocity of a multi-domain mixture is a posterior-weighted convex blend

Let the data be a mixture over N domains (e.g. `geneval`, `ocr`), $q = \sum_{e=1}^{N} \lambda_e\, q_e$, $\lambda_e > 0$, $\sum_e \lambda_e = 1$, each domain e inducing its own path $p_t^{(e)}$ and valid flow $\boldsymbol{u}_t^{(e)}$.

Theorem 1 (Posterior-responsibility decomposition)**.** *The marginal velocity of the mixture teacher is*

$$\boldsymbol{u}_t(\boldsymbol{x}) = \sum_{e=1}^{N} \pi_e(\boldsymbol{x}, t)\, \boldsymbol{u}_t^{(e)}(\boldsymbol{x}), \qquad \pi_e(\boldsymbol{x}, t) = \frac{\lambda_e\, p_t^{(e)}(\boldsymbol{x})}{\sum_{e'} \lambda_{e'}\, p_t^{(e')}(\boldsymbol{x})}, \tag{2}$$

where $\pi_e(\boldsymbol{x}, t) \geq 0$ and $\sum_e \pi_e(\boldsymbol{x}, t) = 1$ is the Bayes posterior that $\boldsymbol{x}_t$ was produced by domain e.

Proof. The mixture path density is $p_t(\boldsymbol{x}) = \sum_e \lambda_e p_t^{(e)}(\boldsymbol{x})$. Introduce the domain indicator E with $P(E = e) = \lambda_e$. By the tower rule applied to (1),

$$\boldsymbol{u}_t(\boldsymbol{x}) = \mathbb{E}[\dot{\boldsymbol{x}}_t \mid \boldsymbol{x}_t = \boldsymbol{x}] = \sum_e P(E = e \mid \boldsymbol{x}_t = \boldsymbol{x})\, \mathbb{E}[\dot{\boldsymbol{x}}_t \mid \boldsymbol{x}_t = \boldsymbol{x}, E = e].$$

The inner conditional expectation is by definition the per-domain marginal velocity $\boldsymbol{u}_t^{(e)}(\boldsymbol{x})$, and Bayes' rule gives $P(E = e \mid \boldsymbol{x}_t = \boldsymbol{x}) = \lambda_e p_t^{(e)}(\boldsymbol{x}) / \sum_{e'} \lambda_{e'} p_t^{(e')}(\boldsymbol{x}) = \pi_e(\boldsymbol{x}, t)$. □

Corollary 1 (Correct MoF form)**.** *A per-sample MoF with N experts, softmax (sum-to-one) gate $w_e(\boldsymbol{x}, t)$ and blend $\hat{\mathbf{u}}_t = \sum_e w_e\, \hat{\mathbf{u}}_t^{(e)}$ can represent the multi-domain teacher* exactly *in the infinite-capacity limit, by matching $\hat{\mathbf{u}}^{(e)} \to \boldsymbol{u}^{(e)}$ (per-domain flow) and $w_e \to \pi_e$ (posterior). Hence the convex constraint $\sum_e w_e = 1$ is not a heuristic: it is the structural form of* (2)*.*

Remark. *Eq.* (2) *is a* convex *combination with input-dependent weights. The intuition "a velocity field is a sum of simpler velocity fields" is precise only in this posterior-convex sense,* ***not*** *as a free additive sum (§3).*

3 Validity of the split: convex vs. free gates

Proposition 1 (When each expert is a valid flow)**.** *Let $\hat{\mathbf{u}}_t = \sum_e w_e(\boldsymbol{x}, t)\, \hat{\mathbf{u}}_t^{(e)}$.*

1. ***Convex,*** $\sum_e w_e = 1$***,*** $w_e \geq 0$ ***(softmax / hard routing):*** *if the experts match the per-domain flows of Thm. 1, each $\hat{\mathbf{u}}^{(e)}$ is individually a valid flow and the blend is the valid mixture flow. Under* hard *routing ($w \in \{0, 1\}$, disjoint domains) the blend is piecewise-single-expert.*

2. ***Free / additive (e.g. independent sigmoid gates,*** $\sum_e w_e \neq 1$***):*** *the blend may fit a strictly larger function class, but an individual summand $\hat{\mathbf{u}}^{(e)}$ need not be a valid flow (it can be a partial/curl-like component that alone does not transport $p_0 \to p_1$), and the blend need not correspond to any single clean probability path.*

Thus **sum-to-one trades capacity for decomposability**: it is the exact structure that keeps "each expert = a valid flow" and makes "split the MoF back into standalone flows" meaningful; free gates buy raw fit at the cost of that interpretation. This is the crux of the "keep vs. drop the convex constraint" design choice.

4 The apparent paradox and its resolution

The informal argument is: (i) big → single is lossy; (ii) big → MoF is lossy but less; (iii) MoF → single is *lossless*; therefore big → MoF → single beats big → single, contradicting a capacity-determined ceiling and the expectation that staging loses performance.

Decompose achieved loss. For a hypothesis class $\mathcal{F}$ and algorithm $\mathcal{A}$, write the achieved loss as

$$\underbrace{\mathrm{Loss}(\mathcal{A}, \mathcal{F})}_{\text{achieved}} = \underbrace{R(\mathcal{F})}_{\text{representational ceiling}} + \underbrace{G(\mathcal{A}, \mathcal{F})}_{\text{optimization gap}}, \qquad R(\mathcal{F}) = \inf_{f \in \mathcal{F}} \mathrm{Loss}(f). \tag{3}$$

Let $\mathcal{F}_C$ be the single-base class (capacity C) and $\mathcal{F}_{\mathrm{MoF}} \supseteq \mathcal{F}_C$ the MoF class.

Proposition 2 (The "lossless" leg is regime-dependent)**.** *MoF → single is (near) lossless* iff *the MoF output function lies in $\mathcal{F}_C$.*

- ***Disjoint-domain / hard routing:*** $\pi_e \in \{0, 1\}$, *so per input only one expert acts; the MoF output is piecewise-single-expert and lies in* $\mathcal{F}_C$ *whenever* C *suffices for the* union *of per-domain functions. Then MoF→single is near-lossless.*

- ***Overlapping domains / soft routing:*** $\pi_e \in (0, 1)$ *forces a genuine blend of two velocities at the* same *input; the output generally lies* outside $\mathcal{F}_C$, *so MoF→single is* ***lossy****.*

Theorem 2 (No capacity is created by staging)**.** *The final student is a single base in* $\mathcal{F}_C$, *hence its achievable loss is bounded below by* $R(\mathcal{F}_C)$ *regardless of the path. In particular* $\mathrm{Loss}\big(\textit{big}\to\textit{MoF}\to\textit{single}\big) \geq R(\mathcal{F}_C)$, *the same representational floor as big→single.*

Corollary 2 (Resolution)**.** *The three premises never hold simultaneously in a way that beats* $R(\mathcal{F}_C)$*:*

- *In the disjoint regime, leg (iii) is ≈lossless, but then leg (ii)'s advantage over big→single is* not *a higher ceiling (a single base of the same per-expert capacity has the same floor on the union); it is an* optimization *advantage (no cross-domain interference during fitting), i.e. a reduction of* G, *not of* R.

- *In the overlapping regime, leg (ii) genuinely lowers* R *(MoF has more capacity), but leg (iii) is then lossy: compressing a* $\mathcal{F}_{MoF}$ *function back into* $\mathcal{F}_C$ *pays exactly the capacity difference.*

Either way, staging cannot beat the $\mathcal{F}_C$ *ceiling. What staging* can *do is shrink the optimization gap* $G(\mathcal{A}, \mathcal{F}_C)$*: the well-optimized MoF is a smoother, easier distillation target (dark knowledge / born-again / progressive distillation), letting the single base approach its* own *ceiling* $R(\mathcal{F}_C)$ *more closely than direct training does. Thus big → MoF → single can beat* direct *big → single* ***without contradiction****: it recovers optimization gap, not representational capacity.*

5 Fixed activated capacity: when does a mixture beat a single base?

Consider top-k-of-N routing: *activated* parameters per input are $\approx kC/1$ (here we take $k{=}1$ so activated $= C$), while *total* parameters are NC.

Proposition 3 (Partitioning gain)**.** *With activated capacity fixed at* C*:*

- *If the teacher velocity is* routable *(input space splits into regions on which the target is simpler, e.g. multi-domain or a nontrivial posterior* $\pi(\boldsymbol{x}, t)$*), then* N *experts each dedicate the full* C *to one region, whereas a single* C *must* share C *across all regions (cross-region interference). The mixture's effective ceiling is the* region-wise *best, which dominates the shared-*C *ceiling:* $R(\mathcal{F}_{MoF,\,k=1}) \leq R(\mathcal{F}_C)$, *often strictly.*

- *If the teacher is* monolithic *(per-input intrinsic complexity* $> C$ *everywhere), each input still sees only* C *activated parameters, so the mixture cannot exceed the single-*C *ceiling.*

Hence a mixture of identical small bases can match a large teacher at constant activated compute *only* to the extent the teacher's velocity is routable/decomposable; this is precisely the structure Thm. 1 exhibits for multi-domain data.

6 Contrast with classic (FFN-level) MoE

	Classic MoE	MoF (this work)
Expert unit	FFN sub-module (not a flow)	full transformer = a valid flow
Routing	per-token, per-layer (fine)	per-sample, output-level (coarse)
Mixing expressivity	high (re-routes each layer)	low (linear blend of final velocities)
Split into standalone flows	no	yes (convex case, Prop. 1)
Param efficiency	high	low (coarse experts)
Matches Thm. 1 form	implicitly, entangled	explicitly ($w_e \approx \pi_e$)

Classic MoE is the stronger tool for *fitting* a teacher at fixed activated compute (Prop. 3); MoF is the right object for the *scientific* question of decomposing a teacher into individually valid, recombinable sub-flows (Cor. 1, Prop. 1). They optimize different objectives: fit efficiency vs. decomposability.

7 Testable predictions

1. **Domain overlap is the master switch.** The more disjoint the domains (posterior $\pi \to \{0, 1\}$, measurable via the router-specialization visualization), the more big$\to$MoF's gain is *optimization* and the closer MoF$\to$single is to lossless; more overlap $\Rightarrow$ real ceiling gain but lossy merge.
2. **Staging beats direct only up to $R(\mathcal{F}_C)$.** big$\to$MoF$\to$single should modestly beat direct big$\to$single, by an amount $\approx$ the direct optimization gap; a *large* win indicates a large direct optimization gap (itself worth study), not broken capacity.
3. **Convex vs. free gate.** Free (sigmoid) gates should reach $\leq$ training MSE(v) of convex (softmax) gates, but the *standalone* sampling quality of an individually extracted expert should be higher for convex (Prop. 1). This separates "fit" from "valid decomposition".
4. **Fair baseline.** A k=1 MoF (activated C) must be compared against a single base of capacity C (equal activated compute), and against a classic MoE of equal activated compute, not against a $2C$ dense model.